Course: Computer Vision

Course Code: ITA0506

Slot: D

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Submitted By

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ITA05-ACTIVITY

SCENARIO BASED ACTIVITY

Mention the techniques used with reason:

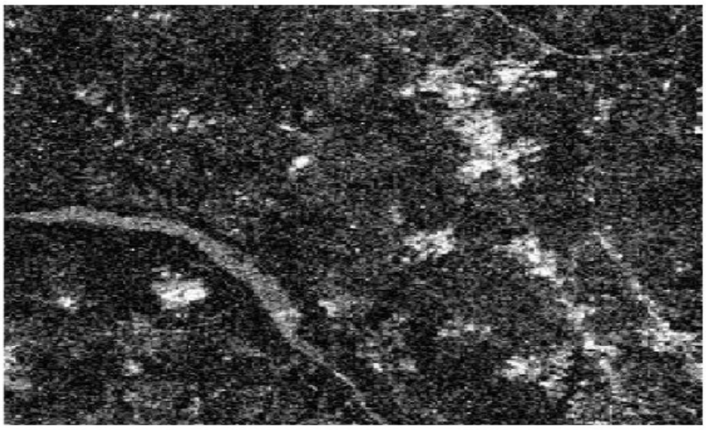
1. **A CCTV image contains salt-and-pepper noise due to transmission errors.**

Scenario 1: CCTV Image Contains Salt-and-Pepper Noise

Technique used: Median Filter (Median Filtering)

Reason: Salt-and-pepper noise appears as randomly scattered black and white pixels in an image. A median filter replaces each noisy pixel with the median value of its neighbouring pixels, which removes the noise while preserving important edges.

2.**A satellite image is corrupted by Gaussian noise from sensors.**

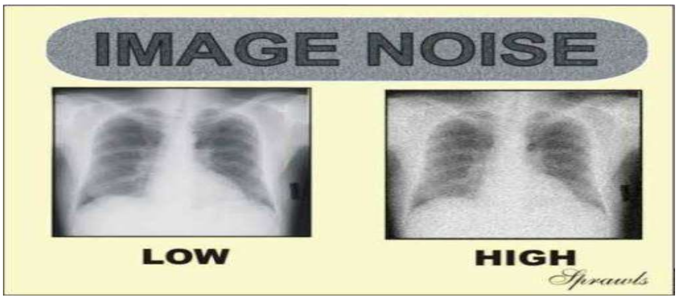


Scenario 2: Satellite Image is Corrupted by Gaussian Noise from Sensors

Technique used: Gaussian Filter (Gaussian Filtering)

Reason: Gaussian noise appears as random variations in pixel intensity, commonly caused by electronic sensors. A Gaussian filter smooths the image by averaging neighbouring pixels using Gaussian weights, which reduces the noise and improves the overall image quality.

3.**A medical X-ray image has random noise but fine edges must be preserved.**

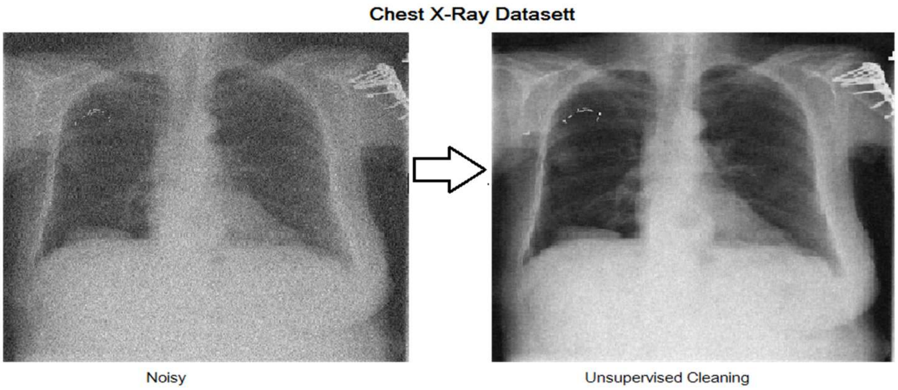


Scenario 3: A Medical X-ray Image Has Random Noise but Fine Edges Must Be Preserved

Technique used: Bilateral Filter (Bilateral Filtering)

Reason: A bilateral filter reduces random noise while preserving important edges and fine details. This is useful for medical X-ray images because the boundaries of bones and internal structures need to remain clear while removing noise.

4.**A scanned document has black and white dots scattered randomly.**

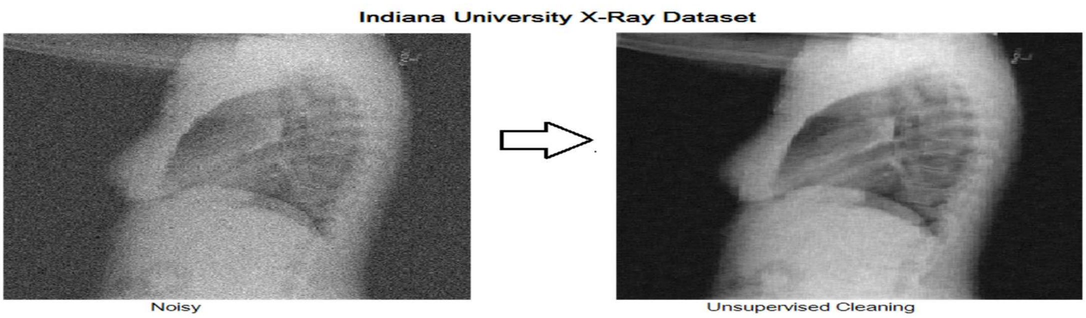


Scenario 4: A Scanned Document Has Black and White Dots Scattered Randomly

Technique used: Median Filter (Median Filtering)

Reason: Random black and white dots are known as **salt-and-pepper noise**. A median filter removes these noisy pixels by replacing them with the median value of neighboring pixels while preserving the edges and clarity of the text.

5.A low-light image shows heavy grain without clear boundaries.



Scenario 5: A Low-Light Image Shows Heavy Grain Without Clear Boundaries

Technique used: Gaussian Filter (Gaussian Smoothing)

Reason: Low-light images often contain random grain or noise. A Gaussian filter smooths the image by averaging neighboring pixel values with Gaussian weights, reducing the grain and producing a cleaner image, especially when there are no clear boundaries that need strong edge preservation.

6.**A foggy road image has very low contrast.**

Scenario 6: A Foggy Road Image Has Very Low Contrast

Technique used: Histogram Equalization

Reason: Fog reduces the difference between bright and dark regions, resulting in low contrast. Histogram equalization redistributes the pixel intensity values over a wider range, improving contrast and making road objects and details more visible.

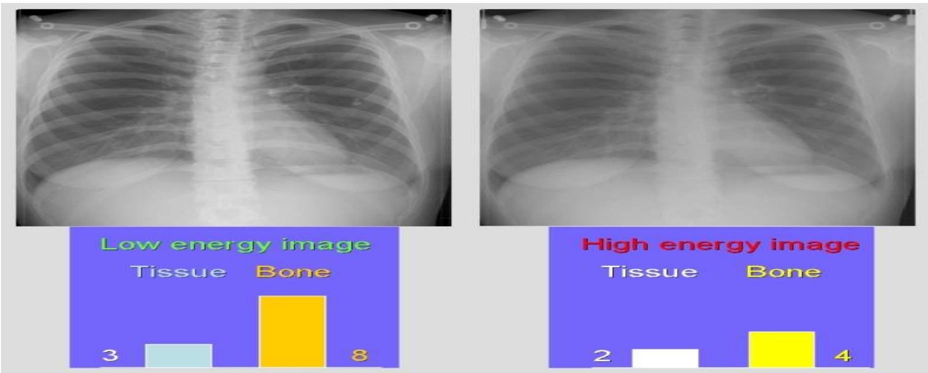
7.An underwater image appears dull and washed out.

Scenario 7: An Underwater Image Appears Dull and Washed Out

Technique used: CLAHE (Contrast Limited Adaptive Histogram Equalization)

Reason: Underwater images often have poor local contrast and unclear details. CLAHE enhances contrast in small regions of the image while limiting excessive enhancement, making underwater objects and details clearer.

8.A chest X-ray shows poor visibility of internal structures.

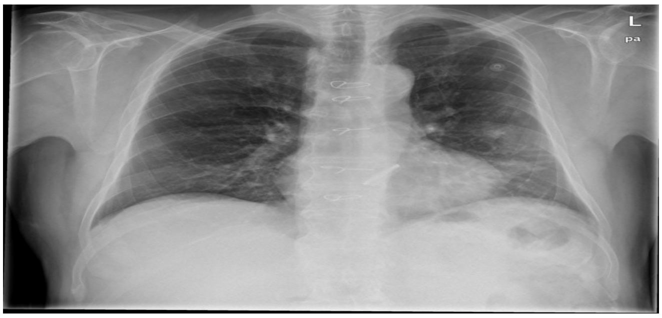


Scenario 8: A Chest X-ray Shows Poor Visibility of Internal Structures

Technique used: CLAHE (Contrast Limited Adaptive Histogram Equalization)

Reason: Chest X-rays may have low local contrast, making internal structures difficult to distinguish. CLAHE enhances contrast in small regions of the image while limiting excessive amplification, making structures and fine details more visible.

9.A night-time traffic image lacks intensity variation.

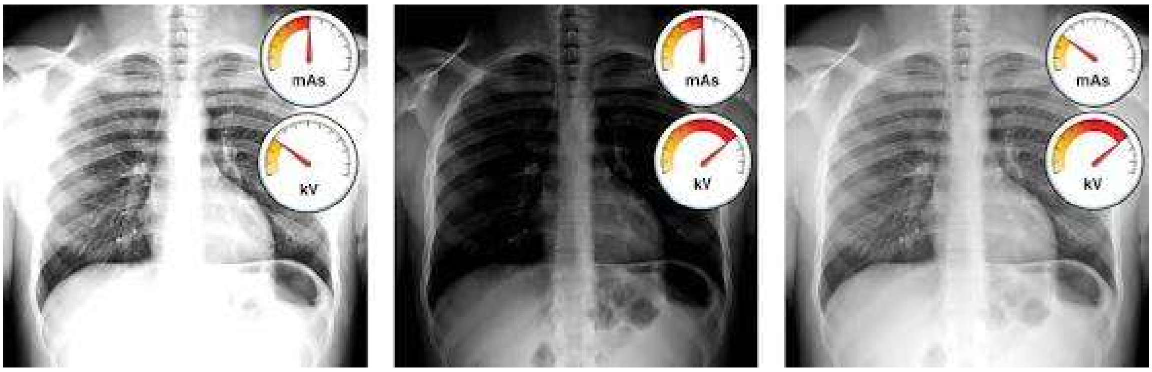


Scenario 9: A Night-Time Traffic Image Lacks Intensity Variation

Technique used: Histogram Equalization

Reason: Night-time images often have poor contrast because most pixel intensities are concentrated in a small range. Histogram equalization spreads the intensity values over a wider range, improving contrast and making vehicles, roads, and other objects more visible.

10.A grayscale image uses only a narrow intensity range.



Scenario 10: A Grayscale Image Uses Only a Narrow Intensity Range

Technique used: Contrast Stretching

Reason: When a grayscale image uses only a narrow range of intensity values, it appears dull and has poor contrast. Contrast stretching expands the intensity values to a wider range, making dark areas darker and bright areas brighter and improving overall image contrast.